

LC 1685 - Sum of Absolute Differences in a sorted array

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Task: For every a_i in nums, make another array b_{rr} , where

$$b_i = \sum |a_i - a_j| \text{ where } j \neq i$$

Example Walkthrough

Given Array: [2, 3, 5]

Then:

- $|2-3| + |2-5| = 1 + 3 = 4$
- $|3-2| + |3-5| = 1 + 2 = 3$
- $|5-2| + |5-3| = 3 + 2 = 5$

Hence, **ans** = [4, 3, 5]

Core Logic Breakdown

Left Contribution logic

Eg: [1, 2, 3, 4]

Then

$$|4-3| + |4-2| + |4-1|$$

$$\rightarrow (4 \times 3) - (1 + 2 + 3)$$

```
arr[idx] += nums[idx] * cnt -  
sum
```

```
and then  
sum += nums[idx]  
cnt ++
```

Right Contribution logic

Eg: [1, 2, 3, 4]

Then

$$|1-2| + |1-3| + |1-4|$$

$$\rightarrow (1 \times 3) - (2 + 3 + 4)$$

but since modded it's like

$$\rightarrow |2-1| + |3-1| + |4-1|$$

$$\rightarrow (2 + 3 + 4) - (1 \times 3)$$

```
arr[idx] += sum - nums[idx] *  
cnt
```

```
sum += nums[idx]  
cnt ++
```

C++ Implementation

```
class Solution { public: vector<int> getSumAbsoluteDifferences(vector<int>& nums)
{ int n = nums.size(); vector<int> ans(n, 0); int sum = 0; int cnt = 0; // left
contri for(int i = 0; i < n; i++) { ans[i] += nums[i] * cnt - sum; sum += nums[i];
cnt++; } sum = 0; cnt = 0; // right contri for(int i = n - 1; i >= 0; i--) {
ans[i] += sum - nums[i] * cnt; sum += nums[i]; cnt++; } return ans; } };
```